25901740 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION April 2026

SECOND SEMESTER

Multi-Disciplinary Courses (MDC) - MG2MDCMBG101 - INDUSTRIAL AND ENTREPRENEURIAL MICROBIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 45 minutes **Maximum Marks: 35**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. Citric acid is mainly produced by [K] / [CO2]

   a). Yeast b). Bacteria

   c). Protozoa d). Clostridium

2. Mushrooms are rich in [U] / [CO4]

   a). Carbohydrates only

   b). Protein and fiber

   c). Cholesterol

   d). Fat only

3. Lactobacillus acidophilus is a [U] / [CO3]

   a). Cyanobacteria

b). Nitrogen fixing bacterium

c). Probiotic bacterium

d). Mould

4. The fastest vinegar production method is [K] / [CO2]

a). Slow process b). Surface fermentation

c). Submerged fermentation d). Tray method

5. Tempeh mould is [K] / [CO2]

a). *Aspergillus niger*

b). *Rhizopus oligosporus*

c). *Bacillus subtilis*

d). *Saccharomyces cerevisiae*

6. The main leavening agent in bread is [U] / [CO2]

a). *Lactobacillus lactis*

b). *Saccharomyces cerevisiae*

c). *Aspergillus niger*

d). *Rhizopus oligosporus*

7. One ethical concern of GM crops is [U] / [CO5]

a). Increased protein

b). Environmental impact

c). Faster growth

d). Better taste

8. Lassi is prepared from [K] / [CO3]

a). Cheese b). Cream

c). Dahi d). Whey

9. A fermenter is also known as a [K] / [CO1]

a). Incubator b). Autoclave

c). Bioreactor d). Laminar hood

10. GM crops are tested under [U] / [CO5]

a). Field trials b). Cooking trials

c). Dry storage d). Packing trials

[1x10 = 10]

Part B

Fill in the blanks questions

Answer all questions

Each question carries **1** marks

11. Pectinase is mainly used for _____ of fruit juices. [U] / [CO2]

12. Spices are added to improve the _____ of sausage [U] / [CO2]

13. The optimum pH for citric acid production is _____. [U] / [CO2]

14. Kefir is prepared using _____ grains. [K] / [CO3]

15. The poisonous mushroom *Amanita phalloides* is commonly known as _____. [U] / [CO4]

[1x5 = 5]

Part C

True or false questions

Answer all questions

Each question carries **1** marks

16. Propionic acid fermentation produces carbon dioxide. [K] / [CO1]

17. Ethanol and carbon dioxide are produced during alcoholic fermentation. [U] / [CO2]

18. Hyphae are thread-like structures of fungi. [K] / [CO4]

19. Butter milk is rich in probiotic bacteria. [K] / [CO3]

20. The Bt gene provides resistance against herbicides. [K] / [CO5]

[1x5 = 5]

Part D

One word / very short answer questions

Answer any **5** questions

Each question carries **1** marks

21. Name one microorganism used in dahi preparation. [K] / [CO3]

22. Name the yeast used in alcoholic fermentation. [K] / [CO1]

23. Define wine. Name the microorganism involved in its production. [K] / [CO2]

24. Identify the first dominant organism in sauerkraut fermentation. [U] / [CO2]

25. Select one medicinal benefits of moderate wine consumption. [U] / [CO2]

26. What type of fermentation is used in protease production ? [K] / [CO2]

27. List any two uses of butter in cooking and baking. [K] / [CO3]

28. Why are probiotic dairy products good for digestion? [K] / [CO3]

[1x5 = 5]

Part E

Short answer questions

Answer any **5** questions

Each question carries **2** marks

29. List any two quality characteristics of good yoghurt. [K] / [CO3]

30. Describe the applications of pectinase in food industry [U] / [CO2]

31. Explain the role of amylase in digestion. [U] / [CO2]

32. Describe the principle and steps involved in vinegar production. [U] / [CO2]

33. List any two microbial sources of cellulase. [K] / [CO2]

34. Discuss the fermentation and maturation stages in beer production. [U] / [CO2]

35. Define cheese and mention the role of rennet in cheese making. [K] / [CO3]

36. What are the end products of alcoholic fermentation? [K] / [CO1]

[2x5 = 10]